

Notes A Functorial Excursion Between Algebraic Geometry and Linear Logic

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1 Basics

Let us start by recalling basic principles of algebraic geometry. The intuition is that a space X comes along with a topological space (X, Σ_X) such that every open set U is equipped with a set $\mathcal{O}_X(U)$ of local operations called *regular functions*. Such regular functions are typically defined as polynomials with division and each $\mathcal{O}_X(U)$ defines a commutative ring. In functorial terms, every space gives rise to a ringed space, that is, a topological space (X, Σ_X) equipped with a contravariant functor $\mathcal{O}_X : \Sigma_X^{op} \rightarrow \mathbf{Ring}$ from the set of opens Σ_X^{op} with the reversed inclusion order. The functorial action of $\mathcal{O}_X$ describes the restriction functions

$$\mathcal{O}_X(V \subseteq U) : \mathcal{O}_X(U) \rightarrow \mathcal{O}_X(V)$$

which takes an operation $f \in \mathcal{O}_X(U)$ defined on an open set U and restrict it to an operation

$$f|_V := \mathcal{O}_X(V \subseteq U)(f)$$

on an open subset $V \subseteq U$. The functorial understanding of the concept of a space is supported by the observation that every commutative ring R defines a ringed space:

$$\mathrm{Spec} R = (\mathrm{Spec} R, \mathcal{O}_{\mathrm{Spec} R})$$

on the set of prime ideals $\mathfrak{p}$ of the commutative ring R . Such a ringed space is called the *affine scheme* associated to R . Every space X of the theory, called

a *scheme*, is a patchwork of affine schemes $\text{Spec } R_i$ glued together using the language of sheaf theory.

Every scheme X comes equipped with a symmetric monoidal category $\mathbf{qcMod}_X$ of quasicoherent sheaf of $\mathcal{O}_X$ -modules. By a presheaf of $\mathcal{O}_X$ -modules M of a ringed space $(X, \mathcal{O}_X)$, we mean a contravariant functor

$$M : \Omega_X^{op} \rightarrow \mathbf{Ab}$$

where $\mathbf{Ab}$ is the category of Abelian groups such that each Abelian group $M(U)$ defines a $\mathcal{O}_X(U)$ -module. The Serre-Swann theorem states a vector bundle on a suitable ringed space $(X, \mathcal{O}_X)$ can be equivalently encoded as its sheaf of sections that defines a locally free and projective sheaf of $\mathcal{O}_X$ -modules. A sheaf of $\mathcal{O}_X$ is called quasicoherent when it is locally presentable, that is, it is locally the cokernel of a morphism of free modules. The category $\mathbf{qcMod}_X$ extends the category of vector bundles to define an Abelian category where kernels and direct images can be computed.

The idea is to associate a model of intuitionistic linear logic, where formulas are understood as generalised vector bundles and proofs are vector fields by building an appropriate linear-non-linear adjunction. To associate a model of linear logic with each scheme X , we take the category $\mathbf{PshMod}_X$ of presheafs of $\mathcal{O}_X$ -modules and their homomorphisms as the underlying symmetric monoidal closed category with tensor product $\otimes_X$, the structural sheaf $\mathcal{O}_X$ as unit, and $-\circ_X$ for internal homs. For the cartesian part of a linear-non-linear adjunction, we take the category $\mathbf{PshCoAlg}_X$ of cocommutative comonoids (or commutative coalgebras) in $\mathbf{PshMod}_X$. It is convenient to the approach based on the functor of points (see [EH00, I.4]). Spec defines a contravariant functor $\text{Spec} : \mathbf{Ring}^{op} \rightarrow \mathbf{Scheme}$ that induces the *nerve functor*

$$\mathbf{nerve} : \mathbf{Scheme} \rightarrow [\mathbf{Ring}, \mathbf{Set}]$$

associating to every commutative ring R the set of R -points in a scheme X defined as $\mathbf{Scheme}(\text{Spec } R, X)$. The nerve functor is fully faithful and it allows one to see every scheme X as a covariant presheaf

$$\mathbf{nerve}(X) : \mathbf{Ring} \rightarrow \mathbf{Set}$$

So we move to schemes to $\mathbf{Ring}$ -spaces and associate a model of intuitionistic linear logic with each $\mathbf{Ring}$ -space X , where formulas are interpreted in $\mathbf{PshMod}_X$ as presheaves of $\mathcal{O}_X$ -module understood as generalised vector bundles on the $\mathbf{Ring}$ -space. The linear-non-linear adjunction (and therefore the exponential modality) is constructed by the Sweedler construction.

2 On the category of rings

Let $(\mathcal{A}, \otimes, \mathbb{1})$ be a symmetric monoidal category with reflexive coequalisers preserved by tensor products in every coordinate. For example, the category $\mathbf{Ab}$ of Abelian groups and homomorphisms. A ring is a monoid object (R, m, e) in

$(\mathcal{A}, \otimes, \mathbb{1})$. In other words, a ring is a triple such that (R, m, e) consisting of an object R and morphisms $m : R \otimes R \rightarrow R$ and $e : \mathbb{1} \rightarrow R$ such that the following diagrams commute:

$$\begin{array}{ccc} R \otimes R \otimes R & \xrightarrow{m \otimes R} & R \otimes R \\ R \otimes m \downarrow & & \downarrow m \\ R \otimes R & \xrightarrow{m} & R \end{array} \quad \begin{array}{ccc} R \otimes R & \xrightarrow{m} & R \\ m \downarrow & \nearrow & \downarrow e \otimes R \\ R & \xrightarrow{R \otimes e} & R \otimes R \end{array}$$

A ring (R, m, e) is commutative is the following triangle commutes:

$$\begin{array}{ccc} R \otimes R & \xrightarrow{\gamma_{R,R}} & R \otimes R \\ & \searrow m & \swarrow m \\ & R & \end{array}$$

A *ring morphism* $u : (R, m_R, e_R) \rightarrow (S, m_S, e_S)$ is a morphism $u : R \rightarrow S$ in the underlying category $\mathcal{A}$ making the following diagrams commute:

$$\begin{array}{ccc} R \otimes R & \xrightarrow{u \otimes u} & S \otimes S \\ m_S \downarrow & & \downarrow m_S \\ R & \xrightarrow{u} & S \end{array} \quad \begin{array}{ccc} & \mathbb{1} & \\ e_R \swarrow & & \searrow e_S \\ R & \xrightarrow{u} & S \end{array}$$

The category **Ring** $(\mathcal{A})$ of rings in $\mathcal{A}$ defined by the tensor product $R, S \mapsto R \otimes S$ in $\mathcal{A}$.

3 On the category of R -modules and modules

Given a ring R in a symmetric monoidal category $\mathcal{A}$, an R -module is a pair (M, act) where M is an object in $\mathcal{A}$ along with a morphism $\text{act} : R \otimes M \rightarrow M$ in $\mathcal{A}$ such that:

$$\begin{array}{ccc} R \otimes R \otimes M & \xrightarrow{m_{R \otimes M}} & R \otimes M \\ R \otimes \text{act} \downarrow & & \downarrow \text{act} \\ R \otimes M & \xrightarrow{\text{act}} & M \end{array} \quad \begin{array}{ccc} & R \otimes M & \\ e_{R \otimes M} \nearrow & & \searrow \text{act} \\ M & \xlongequal{\quad} & M \end{array}$$

Equivalently, an R -module is an Eilenberg-Moore algebra for the monad $A \mapsto R \otimes A$. An R -module homomorphism $f : M \rightarrow N$ between two modules is a map f in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{array}{ccc} R \otimes M & \xrightarrow{R \otimes f} & R \otimes N \\ \text{act}_M \downarrow & & \downarrow \text{act}_N \\ M & \xrightarrow{f} & N \end{array}$$

$\mathbf{Mod}_R$ is the category of R -modules and their morphisms.

A *module* is a pair (R, M) where R is a ring and M is a module. A *module homomorphism* $(u, f) : (R, M) \rightarrow (S, N)$ is a ring morphism $u : R \rightarrow S$ and a morphism $f : M \rightarrow N$ such that

$$\begin{array}{ccc} R \otimes M & \xrightarrow{u \otimes f} & S \otimes N \\ \text{act}_M \downarrow & & \downarrow \text{act}_N \\ M & \xrightarrow{f} & N \end{array}$$

$\mathbf{Mod}$ is the category of modules and their morphisms.

There is a functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ transporting every module (R, M) to the underlying ring R and every morphism (u, f) to u . We will write

$$u : R \rightarrow S \models f : M \rightarrow N$$

for a module morphism $(u, f) : (R, M) \rightarrow (S, N)$. The intuition behind the notation is that every ring homomorphism $u : R \rightarrow S$ induces a “fibre” consisting of all module morphisms $(u, f) : (R, M) \rightarrow (S, N)$ living above and refining the ring morphism $u : R \rightarrow S$. The fibre of πR for a commutative ring is the category $\mathbf{Mod}_R$. The functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ defines a Grothendieck fibration: every pair consisting of a ring morphism $u : R \rightarrow S$ and of an S -module (N, act_N) induces an R -module noted

$$\mathbf{res}_u N = (N, \text{act}'_N)$$

with the action morphism act'_N defined as the composite:

$$R \otimes N \xrightarrow{u \otimes N} S \otimes N \xrightarrow{\text{act}_N} N$$

Moreover, the S -module with a module morphism:

$$u : R \rightarrow S \models \text{id}_N : \mathbf{res}_u N \rightarrow N$$

which is weakly Cartesian in the sense that every module morphism

$$u : R \rightarrow S \models f : M \rightarrow N$$

factors uniquely as

$$M \xrightarrow{(id_R, h)} \mathbf{res}_u N \xrightarrow{(u, id_N)} N$$

for a morphism $h : M \rightarrow \mathbf{res}_u N$ in the category $\mathbf{Mod}_R$. The unique solution h is equal to the original morphism $f : M \rightarrow N$ viewed as a morphism in $\mathbf{Mod}_R$. Given a ring morphism $u : R \rightarrow S$, the existence of a weakly Cartesian map for every S -module N ensures the existence of a functor

$$\mathbf{res}_u : \mathbf{Mod}_S \rightarrow \mathbf{Mod}_R$$

called restriction of scalar along u . Along with the fact that the composite of weakly Cartesian maps is weakly Cartesian we have that the forgetful functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ is a Grothendieck fibration.

Now we use that fact the category $\mathcal{A}$ has reflexive coequalisers, so one can show that $\mathbf{res}_u$ has a left adjoint

$$\mathbf{ext}_u : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_S$$

constructed as follows. Every ring homomorphism $u : R \rightarrow S$ induces an $R \otimes S$ -module notes as $R \otimes_u S$ defined as the reflexive coequaliser of the following diagram

$$\begin{array}{ccc} & \xrightarrow{m_R \otimes S} & \\ R \otimes R \otimes S & \xleftarrow{R \otimes e_R \otimes S} R \otimes S & \\ & \xrightarrow{(R \otimes m_S) \circ (R \otimes u \otimes S)} & \end{array}$$

computed in $\mathcal{A}$. Given three commutative rings R , S_1 and S_2 , we define the functor

$$\otimes_R : \mathbf{Mod}_{S_1 \otimes R} \times \mathbf{Mod}_{R \otimes S_2} \rightarrow \mathbf{Mod}_{S_1 \otimes S_2}$$

The two maps $\text{act}_M : M \otimes R \rightarrow M$ and $\text{act}_N : N \otimes R \rightarrow N$ are deduced by restriction of scalar from the $S_1 \otimes R$ -module structure of M and the $R \otimes S_2$ -module structure of N . The left adjunction functor $\mathbf{ext}_u$ is then defined the following way:

$$\mathbf{ext}_u : M \mapsto M \otimes_R (R \otimes_u S)$$

4 The category $\mathbf{Mod}^\ominus$ of modules and retromorphisms

References

- [EH00] David Eisenbud and Joe Harris. *The geometry of schemes*. Springer, 2000.